MILITARY SPECIFICATION

SHELL, TEST (INERT) 155MM, T5106E2 AND E3 ASSEMBLY

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test shell used as a spotting round for Test Shell PXR6007.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this purchase description to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning methods and pretreatment of ferrous surfaces for organic coatings.

MILITARY

MIL-B-20292 - Blanks, rotating bands for projectile.
MIL-B-20295 - Blanks, centrifugal cost, rotating band.

STANDARDS

FEDERAL

Federal test method standard No. 151 - Metals test methods.

MILITARY

BFBX3-589942 - Pin, Straight, Precision
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

DRAWINGS

ORDNANCE CORPS

75-14-41 - Band, Rotating Gilding Metal
75-14-371D - Grommet Assembly
9203276 - Pin Straight Headless
9233006 - Shell, Test (Inert), 155MM, T5106E2 and E3 Assembly
9233007 - Body
9233008 - Windshield
9233009 - Slug
9233010 - Base
9233011 - Body
ET-9233006 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The shell shall comply with all requirements specified on Drawing (dwg.) 9233006, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Rotating band (see dwgs 9233010 and 75-14-41).

3.5.1 Blanks. The rotating band blanks shall be in the form of a solid ring, annealed if necessary, to comply with MIL-B-20295 when tested as specified in 4.4.1.1.

3.5.2 Physical properties. After removal from base, a 2 inch gage test coupon shall show the following physical properties: Tensile strength of 30,000 pounds per square inch (p.s.i.) and minimum elongation of 25 percent when tested as specified in 4.4.1.2.

3.5.3 Seating. After seating of the band before machining, the diametrical clearance between the band and the seal shall not exceed 0.005 inch at any point of measurement on the band when computed as specified in 4.4.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

3.5.4 Damaged rotating band. Each rotating band on each projectile assembly shall be inspected for the following defects:

a. Cracks

b. Any dent, nick, scratch or gouge which exceeds the allowable depth. (see Fig I).

c. Foreign matter embedded in rotating band material.

Any rotating band which has any of the defects listed in a, b and c shall be rejected and removed from the lot when inspected as specified in 4.4.1.3.

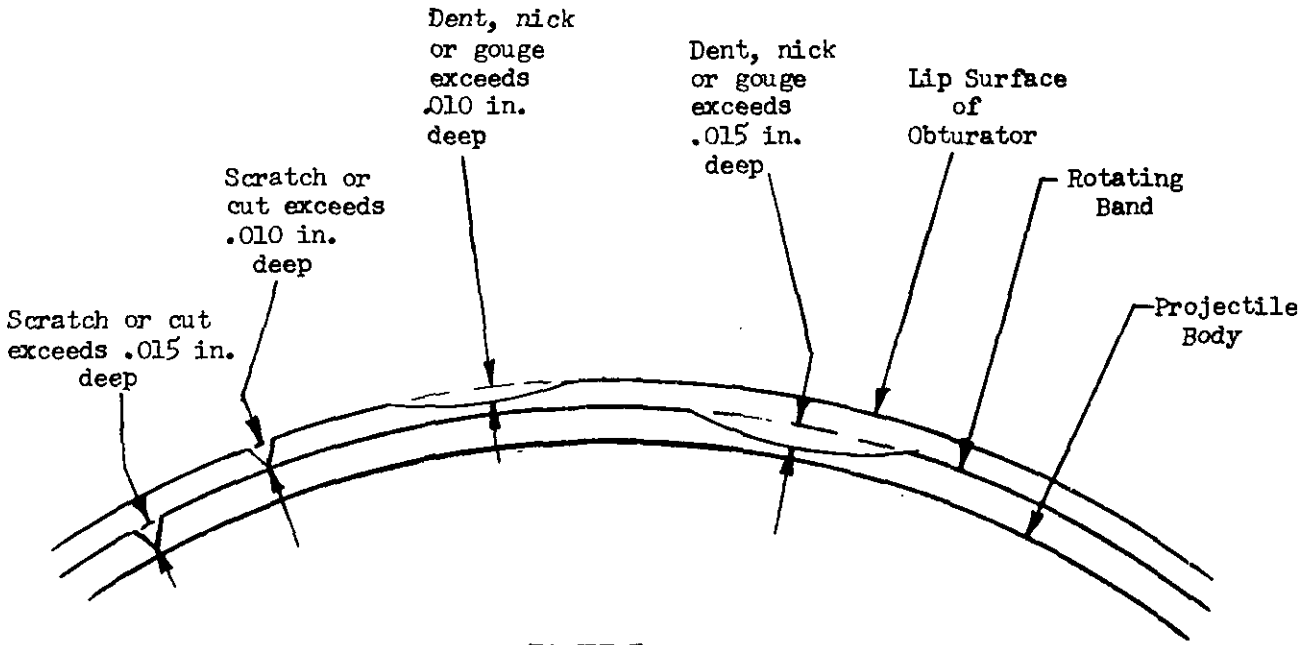


FIGURE I

ROTATING BAND INSPECTION CRITERIA

3.6 Metal to metal contact (see dwg. 9233006).

3.6.1 Joint between base and body. When base (see dwg. 9233010) and body (see dwg. 9233011) are used in the assembly, metal to metal contact shall be considered established when the gap in the joint is calculated in accordance with the drawing. The gap shall not exceed .001 inch or show a variation of more than .0005 inch when measured as specified in 4.4.4.1.

3.6.2 Joint between slug and body. Metal to metal contact between slug and body shall be considered established when the measured length of the slug projection beyond the body equals the calculated length when calculated as specified in 4.4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

3.6.3 Joint between body and windshield. Metal to metal contact between the body and windshield shall be considered established when inspected in accordance with 4.4.4.3.

3.7 Paint adhesion (see dwg. 9233006). Paint adhesion shall comply with Paint Adhesion, section 3, TT-C-490 when tested as specified in 4.4.5.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.5). When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg. 9233006) and 2 sets of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests need not be conducted in order specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Table I

First article inspections and tests

Examination or test	Contractor's inspection before submission	Destructive	Non-Destructive	Parts	Sub-Assembly	Assembly
1. Classification of characteristics (see 4.3.2.1)	X		X	X	X	X
2. Rotating band when sample parts are used (see 3.5)		X		X		X
3. Rotating band when dummy part is used (see 3.5)	X		X	X		X
4. Tensile (see 4.3.3.3)	X		X	X		
5. Charpy (see 4.3.3.2)	X		X	X		
6. Shear (see 4.3.3.4)	X		X	X		
7. Paint adhesion (see 4.3.3.5)	X		X	X	X	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Table III

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	23-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Base (see dwg. 9233010 and 9233007 covering a detail of dwg. 9233006) after rough machining and finish machining of rotating band groove.

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Width of rotating band groove	Gage
102.	Small diameter of rotating band groove	Gage
103.	Projections present, not damaged; correct number on bottom of rotating band groove	Visual
104.	Notches in projections on bottom of rotating band groove present and correctly spaced	Visual
105.	No radius missing	Visual
106.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.8)	Visual

4.3.2.1.2 Base (see dwg. 9233010 covering a detail of dwg. 9233006). This paragraph shall be deleted when dwg. 9233007 is used.

Categories	Characteristics	Method of inspection
Critical:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Major:

101.	Bourrelet diameters	Gage
102.	Pitch diameter of thread	Gage
103.	Major diameter of thread	Gage
104.	Diameter of thread clearance, max	Gage
105.	Width of thread clearance, max	Gage
106.	Length from left face to right side of thread clearance	Gage
107.	Length from left face to rotating band	Gage
108.	Total length	Gage
109.	Length of right outside diameter, max	Gage
110.	Right outside diameter	Gage
111.	Right inside diameter	Gage
112.	Runout of right inside diameter	Gage
113.	Small inside diameter	Gage
114.	Depth of cavity	Gage
115.	Length of right inside diameter	Gage
116.	Run out of right inside shoulder	Gage
117.	63 finish	Visual
118.	Diameter of small grooves in rotating band	Gage
119.	Diameter of large groove in rotating band	Gage
120.	Length of rotating band, min	Gage
121.	Outside diameter of rotating band	Gage
122.	Rotating band (see 3.5.4)	Gage
123.	Contour of rotating band	Visual
124.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 9233010	Visual
125.	Material (see 3.1)	Visual

Minor:

201.	Stamping, metal not raised by stamping	Visual
202.	No radius or chamfer missing	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.8)	Visual-Manual

4.3.2.1.3 Body (see dwg. 9233011 covering a detail of dwg. 9233006). This paragraph shall be deleted when dwg. 9233007 is used.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101.	Left outside diameter	Gage
102.	Length of left outside diameter	Gage
103.	Perpendicularity of left face	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Major:
104. Parallelism of right face.....Gage
105. 63 finish.....Visual
106. Total length.....Gage
107. Run out of inside left diameter.....Gage
108. Inside left diameter.....Gage
109. Middle outside diameter.....Gage
110. Run out of middle outside diameter.....Gage
111. Right inside diameter, max.....Gage
112. Run out of right inside diameter.....Gage
113. Length of right outside diameter, min.....Gage
114. Right outside diameter.....Gage
115. Right diameter to sharp corners, max.....Gage
116. Run out of right outside diameter.....Gage
117. Run out of right inside shoulder.....Gage
118. Length of right inside diameter, max.....Gage
119. Chamfer present.....Visual
120. Objective evidence (see 6.2) available for protective finish in accordance with dwg. 9233011.....Visual
121. Material (see 3.1).....Visual

Minor:
201. Stamping.....Visual
202. Serialization (see 3.4).....Visual
203. Workmanship (see 3.8).....Visual-Manual

4.3.2.1.4 Body (see dwg. 9233007 covering a detail of dwg. 9233006) (see paragraph 4.3.2.1.1). This paragraph shall be deleted when dwgs. 9233010 and 9233011 are used.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Bourrelet diameters.....Gage
102. Pitch diameter of thread.....Gage
103. Major diameter of thread.....Gage
104. Diameter of thread clearance, max.....Gage
105. Width of thread clearance, max.....Gage
106. Length from left face to right side of thread clearance.....Gage
107. Length from left face to rotating band.....Gage
108. Length from rotating band to small outside diameter, min.....Gage
109. Total length.....Gage
110. Length from small outside diameter to right face, min.....Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Major:

111.	Run out of right face	Gage
112.	Small inside diameter	Gage
113.	Run out of small inside diameter	Gage
114.	Small outside diameter, min	Gage
115.	Run out of small outside diameter	Gage
116.	Large inside diameter, max	Gage
117.	Run out of large inside diameter	Gage
118.	Outside diameter at sharp corners, max	Gage
119.	Length of large inside diameter, max	Gage
120.	Run out of right inside shoulder	Gage
121.	Depth of cavity	Gage
122.	63 finish	Visual
123.	Length of rotating band, min	Gage
124.	Diameter of small grooves in rotating band	Gage
125.	Diameter of large groove in rotating band	Gage
126.	Outside diameter of rotating band	Gage
127.	Rotating band (see 3.2.4)	Gage
128.	Contour of rotating band	Visual
129.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 9233007	Visual
130.	Material (see 3.1)	Visual

Minor:

201.	No radius of chamfer missing	Visual
202.	Stamping; metal not raised by stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.8)	Visual-Manual

4.3.2.1.5 Slug (see dwg. 9233009 covering a detail of dwg. 9233006).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101.	Outside diameter, min	Gage
102.	Run out of left and right face	Gage
103.	Total length	Gage
104.	Run out of inside hole	Gage
105.	Inside diameter, max	Gage
106.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 9233009	Visual
107.	Material (see 3.1)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Minor:
201. Stamping.....Visual
202. Serialization (see 3.4).....Visual
203. Workmanship (see 3.8).....Visual-Manual

4.3.2.1.6 Windshield (see dwg. 9233008 covering a detail of dwg. 9233006).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Length of inside left diameter, max.	Gage
102.	Inside left diameter, max.	Gage
103.	Length to left datum diameter	Gage
104.	Length to right datum diameter	Gage
105.	Depth of cavity	Gage
106.	Wall thickness	Gage
107.	63 finish	Visual
108.	Concentricity of nose	Gage
109.	Total length	Gage
110.	Perpendicularity of left face	Gage
111.	Material (see 3.1)	Visual
Minor:		
201.	No radius or chamfer missing	Visual
202.	Stamping; metal not raised by stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.8)	Visual

4.3.2.1.7 Pin, straight headless (see dwg. 9203276 covering a detail of dwg. 9233006). Delete this paragraph when body 9233007 is used.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Diameter	Gage
102.	Length	Gage
103.	Surface finish	Visual
104.	Rockwell hardness	Gage
105.	Chamfer left end	Gage
106.	Chamfer right end	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Minor:
201. Tool makers ink present.....Visual
202. Workmanship (see 3.8).....Visual-Manual

4.3.2.1.8 Grommets, for 155MM 8 Inch and 240MM Projectiles assembly, list of parts and list of spec. and std. (see 75-14-371).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Staples flat against liner.....Visual-Manual

Minor:
201. Workmanship (see 3.8).....Visual

4.3.2.1.9 Shell, test (inert) 155MM, T5106E2 and E3 assembly, (see dwg. 9233006) before assembly of pins.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter of left pin holes (omit when 9233007 body is used).....Gage
102. Depth of left pin holes (omit when 9233007 is used).....Gage
103. Diameter of middle pin holes.....Gage
104. Depth of middle pin holes.....Gage
105. Diameter of right pin holes.....Gage
106. Depth of right pin holes.....Gage
107. Location of middle pin holes.....Gage
108. Location of right pin holes.....Gage
109. Metal to metal contact (see 3.6.1 and 3.6.2).....Gage

Minor:
201. Workmanship (see 3.8).....Visual

4.3.2.1.10 Shell, test (inert) 155MM T5106E2 and E3 assembly (see dwg. 9233006).

Categories	Characteristics	Method of inspection

Critical: None defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

Major:
101. Objective evidence (see 6.2) available for pins in accordance with BFBX3-589942.....Visual
102. Rotating band before assembly of grommet (see 3.5.4).....Gage
103. Weight of shell before grommet assembly.....Gage
104. Grommet assembly present and correctly assembled.....Visual
105. Staking present and correctly spaced on pins.....Visual
106. Pins below flush.....Manual
107. Metal to metal contact (see 3.6.3).....Visual

Minor:
201. Paint missing including bare spot.....Visual
202. Marking.....Visual
203. Serialization (see 3.4).....Visual
204. Workmanship (see 3.8).....Visual-Manual

4.3.2.1.11 Preparation for delivery (see section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Packaging, packing and marking not in accordance with MIL-STD-1169(MU).....Visual

Minor:
201. Workmanship (see 3.8).....Visual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test Classification
a. Rotating band	Major
b. Charpy	Major
c. Tensile	Major
d. Shear	Major
e. Paint adhesion	Minor

Test a is considered destructive when sample parts are used and samples so tested shall not be returned to the lot. The destructive samples are over and above the deliverable quantity of the items required by the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.3.3.1 Rotating band (see dwgs. 9233010 and 9233007).

4.3.3.1.1 Blanks (see 3.5.1). Two samples shall be selected at random from each lot of blanks and tested as specified in 4.4.1. If any sample fails to comply with 3.5.1, the lot of blanks shall be rejected.

4.3.3.1.2 Physical properties and seating (see 3.5.2 and 3.5.3). Prior to machining the rotating bands in the lot, a sample selected at random from each lot of 50 or less produced, or the dummy sample representing the lot, shall be tested in accordance with 4.4.1.2. If the sample fails to comply with 3.5.2 and 3.5.3, the lot of banded bases shall be rejected.

4.3.3.2 Charpy.

4.3.3.2.1 Charpy (see dwgs. 9233009, 9233010, 9233011, 9233007). A sample shall be selected at random from each heat treated lot, prepared and tested in accordance with 4.4.2.1. If any sample fails to comply with drawing requirements, the heat treated lot shall be rejected.

4.3.3.2.2 Charpy (see dwg. 9203276). A sample shall be prepared and tested in accordance with 4.4.2.2. If any sample fails to comply with drawing requirements the heat treated lot shall be rejected.

4.3.3.3 Tensile.

4.3.3.3.1 Tensile (see dwgs. 9233009, 9233010, 9233011, and 9233007). A sample shall be selected at random from each heat treated lot, prepared and tested in accordance with 4.4.2.1. If any sample fails to comply with drawing requirements, the heat treated lot shall be rejected.

4.3.3.3.2 Tensile (see dwg. 9203276). A sample shall be selected at random from each heat treated lot, prepared and tested as specified in 4.4.2.2. If any sample fails to comply with drawing requirements, the heat treated lot shall be rejected.

4.3.3.3.3 Tensile (see dwg. 9233008). A sample shall be selected at random from each heat treated lot, prepared and tested in accordance with 4.4.2.3. If any sample fails to comply with drawing requirements, the heat treated lot shall be rejected.

4.3.3.4 Shear (see dwg. 9203276). One sample shall be selected at random from each heat treated lot and tested as specified in 4.4.3. If any sample fails to comply with drawing requirements, the heat treated lot shall be rejected.

4.3.3.5 Paint adhesion (see 3.7). Each part or assembly shall be tested for paint adhesion as specified in 4.4.5. Any part or assembly which fails to comply with the requirements for paint adhesion in 3.7 shall be classified defective and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-9233006. When either of the conditions listed below exist,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Rotating band (see dwgs. 9233010, 9233007 and 3.5).

4.4.1.1 Blanks (see 3.5.1). The rotating band blanks shall be tested in accordance with MIL-B-20292 or MIL-B-20295.

4.4.1.2 Band seating and physical properties (see 3.5.2 and 3.5.3). Prior to removal the unmachined band shall be measured on three diameters 120 degrees apart. At each of the three diameters measurement along the width of the band shall be taken at stations 1, 2 and 3 totaling 9 measurements in all. Stations 1 and 3 are 5/16 inch from the ends, and Station 2 is at the center. The location of the points of measurement shall be marked on the band along with the orientation of the band with the base. The band shall be removed in a manner that will keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 18 measurements) and the diameter of the band seat at these points shall also be measured. To each of the band seat diameters add the sum of the corresponding two band thicknesses and subtract from the corresponding diameter previously determined. The result will indicate twice the diametrical clearance between the band and its seat as indicated in 3.5.3. If the band is well seated, a negative clearance or interference may be indicated. When irregular band profiles interfere with measurement, the manufacturer is authorized to machine the outside of the band in order to obtain an adequate cylindrical surface. The diameter of the machined band shall exceed the bourrelet diameter. Since a piece must be destroyed to test clearances, a dummy component of similar material will be used for the test providing the dimensions of the band seat is within dwg. 9233006 each time a band is tested. The machine settings used in banding the dummy component or scrap piece shall be the same as the lot being banded. One specimen shall be tested for every 50 pieces or less produced. A maximum of 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

cuts in the rotating band are permissable across the band seat in the test piece. The removed band shall be carefully flattened without hammering and test coupons shall be machined and tested in accordance with Federal Test Method Standard No. 151 except the overall length specified for the specimen shall be determined by the contractor.

4.4.1.3. Damaged rotating band (see 3.5.4). An impression with "cast stone" or (grey stone) shall be taken of each nick, scratch, dent or gouge to determine the depth of the imperfection by a comparator type device or any other accurate device approved by Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NR4. Any other method devised by the contractor to measure the depth of imperfections shall be approved by Picatinny Arsenal, SMUPA-NR4. For allowable depth of the imperfection see Fig. I. In addition, the rotating band shall be visually inspected for any crack or foreign matter embedded in the rotating band. The rotating band shall comply with 3.5.4. This operation shall be witnessed by a Government inspector who will certify that procedures and requirements has been complied with.

4.4.2 Charpy and tensile.

4.4.2.1 Charpy and tensile (see 4.3.3.2.1 and 4.3.3.3.1). The heat treated lot shall not exceed 25 pieces from the same manufacturer and the same mill run of stock. A sample piece, minimum length one inch, shall be selected from the stock at random when preparing the pieces (see applicable drawing) for heat treatment from stock. This sample may be used for both charpy and tensile specimens. The sample shall be included in the heat treated lot and receive the same treatment and process as the heat treated parts. After heat treatment two transverse charpy specimens, when required, shall be selected from the sample 180 degrees apart, with the notch on one specimen facing the outside of the sample and the notch on the other specimen facing the inside of the sample. Two transverse tensile specimens, when required, shall be selected from the sample 180 degrees apart. When solid stock is used the specimens shall be selected from a ring area one inch wide from the outside diameter of the sample. The specimens shall be prepared as specified in Federal Test Method Standard No. 151 using methods 211.1 or 221.1 as applicable and tested on approved test equipment. The average of the test results from the two specimens shall comply with drawing requirements.

4.4.2.2 Charpy or tensile (see 4.3.3.2.2 and 4.3.3.3.2). Two test specimens shall be selected at random from stock for each heat treated lot of parts. These specimens shall be included in the heat treated lot and receive the same treatment and process. After heat treatment, the test specimens shall be prepared and tested on approved equipment in accordance with Federal Test Method Standard No. 151 using methods 211.1 or 222.1 as applicable. Each test specimen shall comply with drawing requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

4.4.2.3 Tensile (see 4.3.3.3). Two test specimens shall be selected 180 degrees apart from a forging selected at random from the heat treated lot. These specimens shall be prepared and tested in accordance with Federal Test Method Standard No. 151, method 211.1. The average of the test results shall comply with drawing requirements.

4.4.3 Shear (see 4.3.3.4). Two suitable test specimens shall be prepared and tested for each heat treated lot on approved test equipment. Each specimen shall comply with drawing requirements.

4.4.4 Metal to metal contact (see 3.6).

4.4.4.1 Joint between base and body (see 3.6.1). The gap between the base and body shall be measured as specified on dwg. 9233006. Inspection records shall be available for each joint until the projectile is fired.

4.4.4.2 Joint between slug and body (see 3.6.2). Measure the depth of counterbore in the body to the nearest .001 inch and record. Measure the length of the slug to the nearest .001 inch and record. Subtract from the length of the slug the length of the counterbore. Measure the length of the slug projection beyond the body of the projectile after assembly of the pins and record.

4.4.4.3 Joint between body and windshield (see 3.6.3). The joint between the body and windshield shall be examined with a 10 power magnifier. Any joint which does not have 50% contact shall be rejected.

4.4.5 Paint adhesion (see 3.7). Paint adhesion shall be tested in accordance with Paint adhesion, section 4, TT-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The shell shall be preserved and packaging in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The shell shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The shell shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Supersession data. This specification includes the requirements of Purchase Description PA-PD-2306B, dated 15 May 1968.

6.5 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

Custodian:
Army - MU

Preparing Activity:
Army - MU

Project No. 1190-A052

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-50291(MU) Shell, Test (Inert) 155mm, T5106E2 and E3 Assembly

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:20Z
Check the source to verify that this is the current version before use.